

**Department of Computer Science and Engineering**  
**University of Dhaka**  
**4th Year 1st Semester Examination, 2025**  
**CSE 4114 - Project work**  
**Project Proposal Form**

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| <b>Title</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Heterogeneous Token Fusion for Transformer-Based Multi-UAV Task Allocation with Deep Reinforcement Learning |
| <p>Multi-UAV systems operating in dynamic environments face a fundamental coordination challenge: efficiently allocating tasks across multiple unmanned aerial vehicles while adapting to real-time changes in task priorities, deadlines, and agent availability. Existing approaches either rely on pure deep reinforcement learning (DRL), which struggles with global coordination and scalability, or apply transformer-based multi-agent methods that treat all entities — UAVs and tasks alike — as structurally identical input tokens, ignoring critical domain-specific distinctions between these fundamentally different entity types.</p> <p>This project proposes a hierarchical coordination framework that decomposes the multi-UAV problem into two levels: a transformer-based global planner responsible for task allocation, and decentralized DRL agents responsible for local execution. The core contribution is a novel Heterogeneous Token Fusion mechanism applied at the transformer input layer. Rather than encoding UAVs and tasks through a shared projection, the proposed architecture employs separate learned projection matrices and type embeddings for each entity class, a priority-aware temporal encoding for task tokens that represents deadline urgency geometrically in the embedding space, and an asymmetric attention masking scheme that enforces semantically meaningful communication constraints between entity types — reducing attention complexity while preserving coordination quality.</p> <p>The expected outcomes of this research include improved task completion rates under dynamic and deadline-constrained scenarios, better scalability as the number of UAVs and tasks increases, and reduced computational overhead compared to standard transformer baselines. The framework will be validated through simulation-based experiments across multiple environment configurations, with ablation studies confirming the independent contribution of each architectural component. This work targets applications in disaster response, surveillance, and time-critical autonomous UAV operations where robust and efficient multi-agent coordination is essential.</p> |                                                                                                             |

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